

Extending the In-Group Advantage: Vocal Emotion Recognition in First and Second Languages by Dutch-English Bilinguals

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Abstract

Previous research finds that vocal emotions are recognized above chance, with higher accuracy typically observed for one's own cultural group—an “in-group” advantage. However, this may not apply uniformly, especially for bilinguals who are highly proficient in a second language (L2) but not members of the L2 cultural in-group. This study investigates Dutch-English bilinguals' recognition of anger, fear, happiness, sadness, and neutral emotion in Dutch (first language; L1), English (L2), and two unfamiliar languages, Arabic and Hindi, using pseudo-utterances. Listeners were highly accurate in recognizing emotions in both Dutch and English. Recognition in Arabic matched their known languages, while Hindi was lower. Additionally, accuracy in English emotion recognition was positively correlated with English proficiency. These results suggest that the in-group advantage can extend to a highly proficient L2, indicating that cultural membership is not strictly necessary for highly accurate vocal emotion recognition.

Keywords: emotion recognition, second language, vocal emotion

Extending the In-Group Advantage: Vocal Emotion Recognition in First and Second Languages by Dutch-English Bilinguals

The expression of emotion is a vital aspect of human communication. Previous research has established that basic emotions such as anger, fear, sadness, joy/happiness, and disgust (Ekman & Friesen, 1971) can be conveyed through behavior, facial expressions, and vocally (Elfenbein & Ambady, 2002). In fact, prosodic and acoustic features such as intonation, pitch, and rhythm can be used to convey emotion through the voice alone (Banse & Scherer, 1996; Juslin & Laukka, 2003; Scherer, 2003), playing a crucial role in social communication by helping listeners interpret a speaker's intention and emotional state. For detailed discussions of the specific acoustic features associated with different emotions and their recognition across languages, see Ekberg et al. (2023), Larrouy-Maestri et al. (2024) and Yildirim et al. (2004).

Research has found that vocal emotion (just as facial emotion) has universal qualities that can be recognized cross-culturally at levels greater than chance (e.g., Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Scherer et al., 2001; Thompson & Balkwill, 2006; Zhang & Pell, 2022; for other perspectives, such as those emphasizing facial emotion recognition being largely context-dependent, see e.g., Barrett, 2011; Barrett et al., 2011; Gendron et al., 2014; Russell & Nelson, 2013). While some aspects of vocal emotion recognition appear to be universal, research additionally indicates that it may be easier to interpret emotional prosody in a listener's native or first language (L1) compared to unfamiliar foreign languages (Elfenbein & Ambady, 2002; Laukka & Elfenbein, 2021; Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Scherer et al., 2001; Thompson & Balkwill, 2006; Zhang & Pell, 2022; but see McCluskey & Albas, 1981; Zhu, 2013). This benefit, known as the *in-group advantage*, is thought to arise due to language or

culture-specific elements of emotional communication aiding in emotion recognition within a cultural community (Elfenbein et al., 2007; Elfenbein & Ambady, 2002).

However, the idea of an in-group advantage may not uniformly apply to all individuals. Bilinguals, for example, speak two languages and may have varying degrees of language proficiency or exposure to the cultural environment of their second language (L2). Little is known about how bilinguals recognize emotions in both languages, especially for a later-learned L2 where they may not belong to the cultural in-group. This study compares emotion recognition in participants' L1, L2, and unfamiliar languages to explore whether the in-group advantage extends to highly proficient bilinguals' L2, despite their lack of in-group membership.

L1 advantage in vocal emotion recognition

Previous studies investigating the recognition of vocal emotions have identified both universal as well as culture-specific aspects of emotion recognition. Across a range of recognition-based task types, languages, and emotions, research has found that listeners can recognize vocal emotion in both their own language and unfamiliar foreign languages at a level greater than chance (Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Sauter, Eisner, Calder, et al., 2010; Scherer et al., 2001; Thompson & Balkwill, 2006; Zhang & Pell, 2022; see Laukka & Elfenbein, 2021 for a recent meta-analysis). Listeners' ability to recognize emotion conveyed through unfamiliar foreign languages supports the "universality" of emotional prosody recognition (Scherer et al., 2001; but note that universality claims have been challenged due to being primarily derived from vocal-based forced-choice recognition studies, for discussion see Gendron et al., 2014; Lorette & Dewaele, 2024). Furthermore, above-chance emotion recognition accuracy has been observed regardless of whether the unfamiliar languages are

similar to the listeners' native language (e.g., within the same language family) or more distantly related (e.g., Pell, Monetta, et al., 2009; Thompson & Balkwill, 2006).

Importantly, while vocal emotion recognition has been found to exceed chance in both a listener's own language and unfamiliar languages, research shows that individuals better recognize vocal emotions from another speaker of their native language (i.e., a member of their own cultural group) compared to recognizing emotions spoken in an unfamiliar foreign language (i.e. cross-cultural emotion recognition; Cho & Dewaele, 2021; Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Sauter, Eisner, Calder, et al., 2010; Thompson & Balkwill, 2006; Zhang & Pell, 2022). This effect has been described as an "in-group advantage" in emotion recognition (Elfenbein & Ambady, 2002). For instance, Thompson and Balkwill (2006) investigated how native English listeners from Canada recognized joy, anger, sadness, and fear in semantically-neutral sentences produced in both their native language (English) and four unfamiliar languages: Chinese, German, Japanese, and Tagalog. Emotion recognition was above chance across all languages, suggesting that some emotional prosodic cues are recognizable across languages, even for languages that the listener does not speak and is not familiar with. However, emotion recognition accuracy was significantly higher for English than for the unfamiliar languages, supporting the in-group advantage. Further supporting this in-group advantage, Laukka et al. (2016) found that even among native speakers of the same language (English), there was an emotion recognition advantage when the speaker was a member of the listener's own cultural group (i.e., Australia, India, Kenya, Singapore, or the United States) as well as differences in emotional production by English speakers from these different cultural backgrounds. Together, these findings highlight the role of shared listener-speaker cultural membership on the in-group advantage, suggesting that it may stem from learned, culturally

specific patterns of emotion production in a culturally-defined group of language users (i.e., theory of cultural learning in emotional behavior, Elfenbein & Ambady, 2002; dialect theory, Elfenbein, 2013; Laukka et al., 2016).

To better compare emotion recognition across known and unfamiliar languages and isolate emotional prosody apart from semantic meaning, studies have used pseudo-utterance stimuli—non-word sentences that follow the grammatical and phonological rules of a particular language (e.g., “*The fector egzullin the boshent*”; Banse & Scherer, 1996; Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Pell & Kotz, 2011; Scherer et al., 2001; see Lorette & Dewaele 2019; 2020; 2024, for studies examining emotion recognition of semantically meaningful stimuli). For instance, Pell, Monetta, et al. (2009) examined how Argentine Spanish listeners identified emotions in pseudo-utterances presented in their native language, Spanish, as well as in three foreign languages they did not speak (English, German, and Arabic). Results showed that emotion recognition was above chance for all language conditions, with higher accuracy in identifying emotions in the listeners’ native language than in the three unfamiliar languages, again supporting the in-group advantage. Within their native language, listeners identified certain emotions more accurately than others, with higher recognition accuracy for joy, anger and fear and lowest accuracy for neutral, disgust, and sadness. This suggests that while the in-group advantage enhances overall recognition in a native language, its effects vary by emotion. Some emotions may be more universally recognizable, whereas others rely more on language-specific prosodic patterns, making them harder to identify in unfamiliar languages.

Much of the existing work on vocal emotion recognition typically compares listeners’ identification of emotions in their native language (where cultural “in-group” membership is inferred implicitly based on the language spoken) with emotion identification in unfamiliar

foreign languages (e.g., Liang et al., 2025; Pell, Monetta, et al., 2009; Sauter, Eisner, Calder, et al., 2010; Scherer et al., 2001; Thompson & Balkwill, 2006). However, the majority of the world's population is bilingual, speaking more than one language. While the “in-group” advantage in emotion recognition is well-replicated, it remains unclear how it extends to L2 users—particularly those who have reached high L2 proficiency but are not themselves members of the L2 cultural environment. Patterns of emotion recognition in an L2 can offer insights into how language familiarity (apart from in-group membership) may contribute to vocal emotion recognition. For instance, highly accurate L2 emotion recognition may suggest that cultural membership is not required for accurate emotion recognition, whereas reduced recognition relative to the L1 may reflect the added influence of broader in-group cultural experience. In this way, bilingualism serves not only as a population of interest but also as a useful context for examining how language proficiency shapes the in-group advantage in vocal emotion recognition.

L2 vocal emotion recognition

Only a small subset of research on vocal emotion recognition has included bilingual participants, and those that do typically test participants in the L2 environment. These studies have replicated the “in-group” advantage effects to varying degrees (Alqarni & Dewaele, 2020; Altrov, 2013; Bhatara et al., 2016; Dromey et al., 2005; Graham et al., 2001; Lorette & Dewaele, 2015; Paulmann & Uskul, 2014; Zhang & Pell, 2022; Zhu, 2013). For example, Altrov (2013) examined recognition of emotion (i.e., anger, joy, sadness, and neutral emotion) in Estonian sentences for three listener groups: Estonians, Russians, and Russian-Estonian bilinguals (living in Estonia). Overall, they observed that the Russian-Estonian group had increased Estonian (L2) emotion recognition accuracy compared to native Russians (living in Russia, with no Estonian

proficiency). The Russian-Estonian group were just as accurate as native Estonians on anger and sadness, but performed more poorly on joy and neutral emotion. These results were interpreted to indicate that Estonian language ability as well as living and working in an Estonian environment improved emotion recognition of Estonian L2 users. This was particularly true for anger and sadness, indicating that certain emotions may be learned by an out-group or L2 user more easily than other emotions.

Similar conclusions were drawn by Paulmann and Uskul (2014), who compared the recognition of emotions (i.e., anger, disgust, fear, happiness, sadness, surprise, and neutral emotion) in English and Chinese for two participant groups: native British English listeners and Chinese-English bilinguals living in the UK. They found in-group advantages for both groups: overall, the native-English listener group was more accurate at recognizing English emotions, while the Chinese-English bilingual listener group was more accurate at recognizing emotions in Mandarin. Importantly, the Chinese-English bilingual listener group performed slightly better on their out-group language (L2 English) than the native English listener group did with their out-group language (unfamiliar Mandarin). The authors suggest that the elevated out-group performance of the bilingual listeners could be attributed to their increased exposure to English while living in their L2 environment.

Zhang and Pell (2022) extended these findings by adding an unfamiliar language (Hindi). Consistent with Paulmann and Uskul (2014), Zhang and Pell (2022) found in-group advantages for both cultural groups—native English listeners performed better in English; Chinese-English bilingual listeners performed better in Mandarin. Additionally, both groups showed higher accuracy in English and Chinese compared to Hindi. Like Paulmann and Uskul (2014) and Altrov (2013), the authors largely interpreted their results on the basis of culture, suggesting that

for the Chinese-English bilingual listeners, more familiarity with English culture may have contributed to their improved recognition accuracy for English stimuli. However, they additionally suggest that their Chinese-English bilinguals' L2 emotion recognition performance may be attributed to their level of English *proficiency*. Greater language proficiency may enhance emotion recognition due to increased familiarity with relevant acoustic-phonetic and prosodic features associated with emotional expression, thereby improving sensitivity to these cues. However, they did not explicitly examine the relationship between emotion recognition and language proficiency in their study.

To our knowledge, only two studies have explicitly attempted to examine the effect of L2 proficiency on L2 vocal emotion recognition (for studies investigating the effect of L2 proficiency on emotion recognition of semantically meaningful stimuli with vocal, audiovisual, and semantic cues, see Lorette & Dewaele, 2019; 2020; 2024). In their study of English language learners living in the US, Graham et al. (2001) found that Japanese-English and Spanish-English bilingual listeners performed more poorly than native English listeners on English emotion recognition. English emotion recognition did not differ for beginner/low-proficiency listeners compared to more advanced/high-proficiency listeners. Similarly, Bhatara et al. (2016) examined the effect of English language proficiency on emotion recognition in French-English bilinguals living in France and found no overall effect of proficiency. Interestingly, recognition of positive emotions (such as interest, pride, relief, and happiness) in the L2 was *negatively* correlated with English proficiency, although the use of real words in the stimuli may have drawn more proficient participants' attention to the lexical items rather than the prosodic features essential for accurately detecting positive emotions.

Overall, results on L2 emotion recognition are mixed. Some studies suggest that bilinguals exhibit better emotion recognition in the L1 than L2, but that L2 emotion recognition may be improved by language or cultural familiarity (Altrov, 2013; Paulmann & Uskul, 2014; Zhang & Pell, 2022). On the other hand, there is little evidence to indicate that this benefit is directly related to differences in L2 proficiency levels (Bhatara et al., 2016; Graham et al., 2001). Moreover, much of the existing literature is limited by unclear definitions of listeners' language backgrounds and a lack of comparisons to performance of first, second, and unfamiliar languages. In addition, most studies (i.e., Bhatara et al., 2016; Graham et al., 2001; Paulmann & Uskul, 2014; Zhang & Pell, 2022) have been conducted in the native cultural environment of the listeners' L2, with bilinguals who have spent considerable time living there, making it difficult to disentangle the impact of L2 proficiency from direct experience with the L2 culture.

Present study

This study examines how the in-group advantage in vocal emotional prosody recognition—typically observed when listeners identify emotions more accurately in their L1 than in unfamiliar foreign languages—extends to a bilingual's second language (L2). We explore the role of language proficiency on emotion recognition by testing listeners who are highly proficient in a second language but are not a member of the L2 cultural in-group. In doing so, this study advances theories of emotion recognition by highlighting the role of language experience without in-group membership. The present study asks two key questions: (a) do in-group advantages in emotion recognition, typically associated with an L1, also extend to a highly proficient L2, even when listeners do not belong to the L2's native cultural in-group? And (b) how does emotion recognition in unfamiliar foreign languages—languages for which the listener

229 neither speaks nor has cultural affiliation with—compare to emotion recognition in the L1 and
230 L2?

231 We address these questions by examining how accurately highly proficient Dutch-
232 English bilinguals recognize four emotions (happiness, sadness, anger, fear) in pseudo-sentences
233 spoken in their L1 (Dutch), L2 (English), and two unfamiliar foreign languages (Arabic and
234 Hindi). Importantly, this participant group is highly proficient in L2 English and has regular
235 exposure to nonnative English as part of daily life in the Netherlands (L1 Dutch environment).
236 Additionally, participants were tested in their L1 cultural setting and had little to no prior direct
237 immersion in a native English-speaking cultural environment (i.e., were not part of a native
238 English-speaking cultural “in-group”).

239 Building on the theoretical frameworks reviewed in the Introduction, three predictions
240 can be outlined. Prediction (a), which we call the *language-proficiency advantage hypothesis*,
241 predicts that the in-group advantage extends to the highly proficient L2, resulting in similar
242 emotion recognition in L1 and L2, with listeners performing better in the two languages they
243 speak than in the two unfamiliar foreign languages. If the *language-proficiency advantage*
244 *hypothesis* is supported, it would imply that highly proficient bilinguals are as effective at
245 detecting emotions in their later-learned L2 as in their L1—even in the absence of in-group
246 membership in the L2 environment—and would show a clear advantage for both languages they
247 speak over unfamiliar foreign languages.

248 Prediction (b), which we call the *cultural-membership hypothesis*, implies that the in-
249 group advantage *does not extend* to the L2, leading to greater emotion recognition in the L1
250 compared to the L2. This hypothesis ascribes an important role to cultural membership for
251 developing emotion recognition skills comparable to those in the L1. In this case, emotion

recognition performance in the L2 would be similar to that in the unfamiliar foreign languages, supporting the notion that proficiency alone—without native cultural experience—does not facilitate the same level of emotion recognition as in the L1.

Prediction (c), which we call the *L2 modulation hypothesis*, suggests that the L2 proficiency modulates the in-group advantage, such that emotion recognition is highest in the L1, followed by the L2, with the lowest performance observed in the unfamiliar languages. This would indicate that being a member of the cultural in-group supports superior recognition accuracy for the L1. High L2 proficiency would offer a moderate advantage over unfamiliar languages, albeit less than that of the L1.

Methods

Participants

Fifty-seven Dutch-English bilinguals were recruited from Radboud University in Nijmegen, the Netherlands. Data from nine participants was discarded: one due to not following task instructions, one due to reporting English as their most proficient language, three due to exceeding the intended age range, and four due to living in an English-speaking environment for 10 or more years. The final sample included 48 participants (27 women, 15 men, 6 nonbinary; mean age = 22.75 years, min = 18, max = 40). Our target sample size was determined based on previous studies on emotion recognition using similar experimental designs and our final sample size is comparable to that of similar studies in this field (i.e., Altrov et al., 2013; Bhatara et al., 2016; Cho & Dewaele, 2021; Liang et al., 2025).

All included participants spoke Dutch as their first language and learned English as a second language (mean age of English acquisition = 8.92 years; min = 2, max = 12), in a school setting. Only two participants can be considered simultaneous bilinguals, reporting L2

acquisition before the age of four; the rest are sequential bilinguals¹. Participants were highly proficient English speakers with an average score on the LexTALE English lexical decision task of 85.6 ($SD = 10.69$; max score is 100). The average self-rated speaking proficiency for Dutch (9.67, $SD = 0.56$; max score = 10) was greater than the average self-rated speaking proficiency for English (7.54, $SD = 0.97$; $t(47) = -12.9$, $p < .001$). Furthermore, participants reported how many years they have spent living in English- and Dutch-speaking environments (categorical response options ranged from “zero years” to “ten or more years” in one-year increments). All participants reported that they had spent ten or more years living in a Dutch-speaking country. Most participants reported spending little to no time living in an English-speaking country (35 spent zero years, 11 spent one year or less, and 2 spent one to two years).

Participants had no or limited experience speaking or listening to Arabic or Hindi. As self-reported on a language history questionnaire, 47 participants had no experience with Hindi (1 reported having traveled to India). 42 participants had no experience with Arabic (with 6 reporting exposure through movies/TV/music/pop culture or having an Arabic speaking friend). Prior to testing, all participants provided informed consent, as approved by the Institutional Review Board at The Pennsylvania State University and the Humanities Ethics Review Board at Radboud University. Participants were compensated at a rate of 10 EUR per hour for their participation.

Materials

Materials for this study consisted of Dutch, English, Hindi, Arabic pseudo-sentence recordings, spoken with *angry*, *sad*, *happy*, *fearful*, and *neutral* emotion. Critically, all experimental stimuli were pseudo-sentences in which content words were replaced by

¹ We conducted an exploratory analysis excluding the two simultaneous bilingual participants. The overall pattern of results were unchanged. The results of these analyses are available on our OSF repository.

semantically meaningless but phonotactically legal words (for example in English, *The fector egzullin the boshent*). This format retains phonological and some grammatical properties of the language, allowing speakers to easily produce each sentence with different emotional inflections. However, critical to the experimental design of the present study, all words were semantically meaningless, thus, there was no risk of participants accessing semantic information or emotional associations, and decisions were made purely on prosodic cues.

The English, Hindi, and Arabic stimuli used in this study were carefully created and validated by Pell, Paulmann, et al. (2009). For each language, pseudo-sentences were first constructed by a native speaker of each language. Then these pseudo-sentences were produced and recorded by native speakers of each language (English speakers from Quebec/Ontario, Hindi speakers from Central India, and Jordanian/Syrian dialect Arabic speakers from the Middle East) using the same emotional elicitation procedure (see Pell, Paulmann, et al., 2009). Each speaker produced each pseudo-sentence with different emotional inflections including anger, disgust, fear, sadness, happiness, pleasant surprise, and neutral emotion. The stimuli for each language were then entered into perceptual rating studies, where they were rated by native speakers of that language. For full details on the stimuli creation, emotional elicitation and recording procedures, and results from the perceptual rating studies for the English, Hindi, and Arabic stimuli sets, see Pell, Paulmann, et al. (2009).

Dutch stimuli creation

The Dutch stimuli were created by carefully following the methods outlined in Pell, Paulmann, et al. (2009), using the same approach as was used for creation and validation of the English, Arabic, and Hindi stimuli sets. A native Dutch speaker (author JGV) created a list of 30 Dutch pseudo-sentences, which were phonotactically and syntactically legal in Dutch, but in

which all content words (nouns and verbs) were replaced by semantically anomalous pseudo-words (e.g., *Hij belieerde een parf*). On average, the sentences were 4.5 words long (min = 3, max = 6). Two female native speakers of Dutch (ages 21 and 24), not affiliated with or familiar with the nature and goals of the study, recorded the pseudo-sentences in four emotional inflections: anger, fear, happiness, and sadness, and non-emotional neutral fillers. In line with the procedure used by Pell, Paulmann, et al., (2009), before recording the pseudo-sentences, the speakers practiced eliciting each emotion with lexicalized Dutch sentences to facilitate later production of pseudo-sentences in a way that accurately reflected natural prosodic inflections for each emotion. While recording the pseudo-sentence stimuli, speakers were encouraged to speak naturally, avoiding exaggeration. Each sentence was repeated three times, with all sentences for one emotion recorded before proceeding to the next emotion. The order of emotions was randomized between speakers. Recordings were conducted with a digital recorder (TASCAM DR-100 MK III) and an external microphone (STAGG IMP-600 Dynamic Microphone). Sentences were segmented into individual 16-bit WAV files and adjusted to a root-mean-square amplitude of 70dB sound pressure level. The sampling frequency was set at 44.1 kHz using Praat software (Boersma & Weenink, 2022). Dutch stimuli created for this experiment are available on OSF.

Dutch stimuli validation

A validation study was conducted with 30 native Dutch speakers (21 male; 9 female; mean age = 28.53, $SD = 9.94$) all residing in the Netherlands, to identify the most accurate stimuli for use in the current experiment. Participants (none of whom participated in the main experiment) were recruited via Prolific[©] and completed the study online via Qualtrics (Qualtrics, Provo, UT). They listened to all pseudo-sentences produced by both speakers in a randomized order and answered a multiple-choice question indicating which emotion they thought the

speaker was expressing: angry, fearful, happy, sad, or neutral. Data were then used to calculate the average accuracy for each sentence.

Stimuli selection

Perceptual data from the Dutch validation study, along with the validation data for the English, Arabic, and Hindi stimuli sets from Pell, Paulmann, et al. (2009), were used to select the final sentences for this experiment. The highest-rated female speaker of each language was chosen as the exemplar speaker for that language. For each emotion, the top 12 most accurately rated sentences for each speaker were selected, resulting in 60 sentences per speaker. See Table 1 for validation study accuracy ratings for each language and emotion. Validation study accuracy rates are controlled for in the statistical analysis, to account for potential baseline stimuli differences (see Accuracy Analysis below).

< Insert Table 1 here >

Experimental procedure

Experimental procedures closely followed those of Pell, Monetta, et al. (2009). Participants were tested individually in sound-attenuated booths. Participants listened to each sentence presented one at a time; after each sentence, they were asked to judge how the speaker feels based on their voice by choosing one of five options (angry, fearful, happy, sad, or neutral) using a mouse click. The experiment was blocked by language. For each language, participants first listened and responded to all 60 sentences in quiet and then the same sentences with background babble. Only the quiet listening conditions are reported here. The order of the languages was counterbalanced across participants and the sentences were randomized. Prior to the main task, participants completed a practice session with five sentences per speaker (one for each emotion), with two in quiet and three in background babble.

The experiment was programmed and presented with E-Prime 3.0 software (Psychological Software Tools, Pittsburg, PA). Stimuli were delivered through headphones (Edifier Hi-Fi-Stereo H840), with volume set at a consistent level between participants. Accuracy of responses were recorded throughout the experiment.

After the experiment, participants completed a series of questionnaires and tasks. First, they completed a debriefing survey related to the emotion recognition experiment. This survey was designed to gather feedback on which speakers and emotions the participants found easiest and most challenging to recognize. They then completed the LexTALE lexical decision task to measure receptive vocabulary in L2 English (Lemhöfer & Broersma, 2012). Lastly, they completed a language history questionnaire and the Schutte Self-Report Emotional Intelligence Test (Schutte et al., 1998) to measure emotional intelligence.

Accuracy analysis

Data analyses were conducted in R (Version 4.2.2, R Core Team, 2022). Generalized linear mixed-effects models with a binomial family function were fitted to behavioral accuracy data (1 = accurate, 0 = inaccurate) with the *lme4* package (Bates et al., 2015). Language (Dutch, English, Arabic, Hindi), Emotion (anger, fear, sadness, happiness), and their interaction were included as fixed effects. Item and Participant were included as random intercepts. Language and Emotion were [coded using sum-to-zero contrasts](#). Note that non-emotional neutral filler stimuli were excluded from the statistical analyses, as in Pell, Monetta, et al. (2009). Additionally, although care was taken to include the most highly rated stimuli in the experiment, the validation ratings for the stimuli differed between the four languages. Thus, mean-centered validation rating for each item was included as a covariate. Model fitting began with random by-participant slopes for language and emotion. Non-convergence was addressed by iteratively simplifying the

random effects structure, first by removing the interaction between the random slopes, then removing the individual random slope for emotion, then for language. The final model reflects the maximally-supported model (Barr et al., 2013). To test the statistical significance of the fixed effects, type-III analysis-of-deviance tables were calculated, and Wald chi-square tests were conducted with the *car* package (Fox & Weisberg, 2019). To evaluate the simple effects of each fixed effect, estimated marginal means were calculated and pairwise comparisons were conducted with the *emmeans* package (Lenth, 2024), and pairwise p-values were adjusted with the Tukey method.

Results

Overall, emotion recognition rates were high, and the target response was the most frequently selected response for each emotion and language combination. To more closely examine how listeners classified each emotion within each language, a stimulus-response confusion matrix was calculated from the responses, as seen in Figure 1. The resulting matrices revealed clear response patterns, indicating that classification was not random. Listeners demonstrated high overall accuracy for recognizing emotions (81.15%; over 4x 20% chance accuracy) and high accuracy for each language condition (all $ps < .0001$, calculated using *emmeans* from the model below).

< Insert Figure 1 here >

We observed significant effects of Language ($\chi^2(3) = 50.89, p < .001$), Emotion ($\chi^2(3) = 18.64, p < .001$), and their interaction ($\chi^2(9) = 47.39, p < .001$) on accuracy. To probe the two-way interaction between Language and Emotion, we compared the levels of Language within each level of Emotion as shown in Figure 2.

For *angry* stimuli, predicted accuracies for Dutch ($M = 0.95$, 95% CI [0.89, 0.97], $p = .003$), English ($M = 0.95$, 95% CI [0.92, 0.97], $p = .001$), and Arabic ($M = 0.94$, 95% CI [0.9, 0.97], $p = .003$) were greater than for Hindi ($M = 0.8$, 95% CI [0.69, 0.88]). Dutch, English, and Arabic accuracies did not differ ($ps > .05$). For *fearful* stimuli, predicted accuracy for Arabic ($M = 0.97$, 95% CI [0.95, 0.99]) was greater than for Dutch ($M = 0.85$, 95% CI [0.77, 0.91], $p < .001$) or Hindi ($M = 0.77$, 95% CI [0.57, 0.89], $p < .001$), which did not differ from each other. Predicted accuracies for English ($M = 0.93$, 95% CI [0.87, 0.96]) did not differ from the other three languages ($ps > .05$). For *sad* stimuli, predicted accuracies for Dutch ($M = 0.9$, 95% CI [0.83, 0.95], $p = .006$) and English ($M = 0.95$, 95% CI [0.89, 0.97], $p < .001$) were greater than for Hindi ($M = 0.72$, 95% CI [0.57, 0.83]). Accuracy for English was greater than Arabic ($M = 0.81$, 95% CI [0.71, 0.89], $p = .04$) which did not differ from the other two languages ($ps > .05$). Predicted accuracies for *happy* stimuli did not differ between the four languages ($ps > .05$).

We also probed the Language-by-Emotion interaction by comparing levels of Emotion within each level of Language. Within Dutch stimuli, predicted accuracy for *anger* was greater than for *happiness* ($p = .03$). Predicted accuracy for *sad* and *fear* did not differ from each other or from the other emotions within Dutch stimuli. Within English stimuli, predicted accuracy for *anger*, *sadness*, and *fear* were greater than for *happiness* ($M = 0.81$, 95% CI [0.69, 0.89]; $p = .008$, $p = .007$, $p = .03$), but did not differ from one another ($ps > .05$). Within Arabic stimuli, predicted accuracy for *anger* and *fear* were greater than for *sadness* ($p = .005$, $p < .001$), and predicted accuracy for *fear* was also greater than for *happiness* ($p = .02$). None of the other comparisons within Arabic stimuli was significant ($p < .001$). Within Hindi stimuli there were no significant differences between conditions ($ps > .05$).

To follow up on the main effects of Language and Emotion, we investigated their simple effects. For Language, averaging across levels of Emotion, predicted accuracies for Dutch ($M = 0.89$, 95% CI [0.85, 0.92], $p = .002$), English ($M = 0.92$, 95% CI [0.89, 0.95], $p < .001$), and Arabic ($M = 0.93$, 95% CI [0.89, 0.95], $p < .001$) were significantly higher than for Hindi ($M = 0.79$, 95% CI [0.73, 0.84]). No other pairwise comparisons within Language were significant ($ps > .05$). For Emotion, averaging across levels of Language, predicted accuracies were greater for *angry* ($M = 0.93$, 95% CI [0.9, 0.95]) than for *sad* ($M = 0.87$, 95% CI [0.82, 0.91]; $p = .02$) or *happy* ($M = 0.85$, 95% CI [0.81, 0.89]; $p = .003$) stimuli. Predicted accuracies were greater for *fearful* ($M = 0.91$, 95% CI [0.87, 0.94]) than for *happy* ($p = .04$) stimuli. No other pairwise comparisons within Emotion were significant ($ps > .05$).

In addition to the effects of Language and Emotion, there was a significant effect of validation rating such that an increase in validation rating was associated with an increase in accuracy, ($\chi^2(1) = 13.33$, $p < .001$).

< Insert Figure 2 here >

Exploratory correlations with language proficiency measures

To further explore the relationship between listeners' emotion recognition and language ability, we calculated Pearson's correlations between raw emotion recognition accuracy for L1 Dutch and L2 English and various language measures, including English LexTALE scores, self-rated speaking proficiency in both Dutch and English, and English age of acquisition. We corrected for multiple correlations using Benjamini–Hochberg correction for false discovery rate. For simplicity, only significant correlations between emotion recognition accuracy and language measures will be reported here, with the full correlation matrix shown in Figure 3. English

LexTALE scores were positively correlated with English emotion recognition accuracy ($r = 0.38$, $p = .008$, $p_{adj} = .04$).

< Insert Figure 3 here >

Subjective ratings of emotion recognition difficulty

In a debriefing questionnaire, participants were asked to rate their perceived difficulty of recognizing emotions in each language condition on a scale of 1 (extremely easy) to 5 (extremely difficult). The percentage of participants who selected each response for each language is presented in Table 2. For both English and Dutch stimuli, the majority of participants indicated that emotions were somewhat *easy* to recognize. In contrast, for Arabic and Hindi stimuli, the majority of participants reported that emotions were somewhat *difficult* to recognize.

< Insert Table 2 here >

Discussion

This study examined to what extent the commonly found “in-group” advantage in emotion recognition (i.e., greater accuracy for emotion recognition in the L1 compared to foreign languages) extends to a bilingual’s L2, in bilinguals who are not immersed in their L2 and are not a member of the cultural in-group. Additionally, we explored to what extent emotion recognition for unfamiliar foreign languages—languages for which the listener neither speaks nor has cultural affiliation with—compared to recognition in the L1 and L2. To do this, we examined Dutch-English bilinguals’ recognition of vocal anger, sadness, fear, happiness, and neutral emotion in pseudo-utterances in their L1 (Dutch), highly proficient L2 (English), and two unfamiliar languages (Hindi and Arabic).

This study tested three hypothesized outcomes: the *language-proficiency advantage hypothesis* in which the in-group advantage extends to the highly proficient L2, resulting in similar L1 and L2 emotion recognition with both exceeding unfamiliar languages; the *cultural-membership hypothesis* in which the in-group advantage does not extend to the L2 due to lack of L2 cultural in-group membership, resulting in L2 accuracy similar to unfamiliar accuracy; or the *L2 modulation hypothesis* in which the L2 modulates the in-group advantage, such that emotion recognition is highest in the L1, followed by the L2, with the lowest performance observed in the unfamiliar languages. The findings are discussed in detail below.

Universality of emotion recognition

Overall, listeners' accuracy for all languages and emotions was high, exceeding chance accuracy by at least three times. These findings replicate previous research demonstrating high accuracy in emotion recognition for both familiar and unfamiliar languages (i.e., Elfenbein & Ambady, 2002; Laukka & Elfenbein, 2021; Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Scherer et al., 2001; Thompson & Balkwill, 2006; Zhang & Pell, 2022). Given that our stimuli were pseudo-utterances—with no linguistic or semantic information—listeners demonstrated the ability to recognize emotion across both known and unfamiliar languages from prosodic features alone, supporting the universality of vocal emotion recognition (Elfenbein & Ambady, 2002; Scherer et al., 2001).

Findings on L2 emotion recognition

As expected, listeners were highly accurate at identifying emotions in their L1, Dutch. Listeners were also highly accurate at recognizing emotions in their L2, English. In fact, we did not find significant differences in emotion recognition accuracy between L1 Dutch and L2 English overall, or for any specific emotions. These results support the *language-proficiency*

advantage hypothesis—we find the in-group advantage for emotion recognition extends to a highly proficient L2.

Critically, these findings suggest that even bilinguals living and tested in their L1 cultural environment (who are not members of the L2 cultural in-group, and the majority of whom have spent no time living in the L2 native cultural environment) can learn to recognize emotion in their L2 at a level comparable to emotion recognition in their L1. These results inform existing theories of emotion recognition such as the theory of cultural learning in emotional behavior (Elfenbein & Ambady, 2002) and the dialect theory (Elfenbein et al., 2007; Laukka et al., 2016), which emphasize the need for cultural specific aspects of emotional behavior to be learned via direct cultural membership.

Our results suggest, at least for highly proficient L1 Dutch speakers of L2 English who learned their L2 in a school setting, that cultural in-group membership is *not* a prerequisite for achieving native-like levels of emotion recognition and point to language proficiency as an important factor. For highly proficient L2 listeners, interactions from classroom learning and engagement with other L2 English or native English speech (e.g., via social media or television) in their L1 Dutch environment are sufficient for learning the acoustic-prosodic cues necessary for emotion recognition, even without L2 cultural membership. This conclusion aligns with research which emphasizes that vocal expressions of emotion can be processed without extensive cultural learning (Sauter, Eisner, Ekman, et al., 2010). In other words, listeners can extract and interpret emotional prosodic cues across languages, with greater language proficiency promoting greater basic emotion recognition accuracy in that language. Consequently, bilinguals without L2 cultural membership are still able to learn and extract key acoustic-prosodic cues through

language learning and interaction alone, achieving recognition performance comparable to that of their L1.

Further supporting this role of language proficiency as a crucial factor in L2 emotion recognition, we find significant positive correlations between English accuracy and language proficiency (the LexTALE test of vocabulary knowledge). While previous studies (Bhatara et al., 2016; Graham et al., 2001) did not find significant correlations between L2 proficiency and emotion recognition, here we tested highly proficient L2 users (rather than lower-level L2 learners as in Graham et al., 2001) and used semantically meaningless pseudo sentences (eliminating potential negative effects of proficiency as seen in Bhatara et al., 2016 who used stimuli with real words). Positive correlations found here suggest that L2 emotion recognition can be modulated by language proficiency, giving further evidence that L2 prosodic cues can be learned without in-group cultural membership.

It is important to note, however, that it is difficult to fully disentangle language familiarity with cultural *exposure*. More specifically, in the L1 environment of the present study's listener group (the Netherlands), English is widely used as a second language. A large majority of the population (approximately ~90%) is proficient in English (European Commission, 2012). This high level of proficiency is reflected in the country's high ranking in the English Proficiency Index (Education First, 2024), scoring above the global average. Within the country, English is commonly used in education, business, and tourism, and many Dutch people are fluent in English, especially in urban areas and among younger generations (Edwards, 2016). Although our participants are not "L2 English in-group members," they likely have substantial exposure to English-speaking individuals or communities through various channels such as academic settings, interactions with other L2 or L1 English speakers, and via virtual

channels such as English-language content on the internet, non-dubbed films, online gaming, or social media. L2 language classes themselves often include authentic materials from English-speaking countries, and expose students to cultural “mandates” or “scripts” (i.e., Goddard & Wierzbicka, 2004; Mesquita et al., 2017). As a result, participants’ proficiency in English most likely also provides them with considerable cultural context for successful emotion recognition in their L2, and this cultural exposure likely increases with increased language proficiency.

Disentangling language proficiency and cultural exposure is empirically difficult, particularly in today’s highly globalized, interconnected world. “In-group” membership as typically described (i.e., Elfenbein et al., 2007; Elfenbein & Ambady, 2002) may no longer be a relevant categorization as individuals around the world are increasingly exposed to other cultures virtually via easily accessible content on the internet and social media. Nonetheless, future research could attempt to more directly examine the differential effects of language proficiency versus cultural exposure in at least two ways. First, studies could include a questionnaire to characterize participants’ prior cultural exposure or cultural characteristics (for an example, see De Groot et al., 2021). Second, future studies could explicitly compare a group learning an L2 with no or little exposure to the native culture versus a group learning an L2 with exposure to or immersion in a native cultural environment to better isolate the effects of language versus culture (see for example, Hess et al., 2013).

Findings on unfamiliar language emotion recognition

Further supporting the *language-proficiency advantage hypothesis*, we find an overall advantage for both Dutch and English compared to Hindi. This effect replicates in-group advantages found by Zhang and Pell (2022), who found that Chinese-English bilinguals immersed in their L2 environment were better at emotion recognition in both their L1 and L2

compared to Hindi as a foreign language. Specifically, we find that this effect was driven by negative emotions, *anger* and *sadness*. Interestingly, although Arabic and Hindi are both unfamiliar languages to our Dutch-English bilingual participants, emotion recognition was more accurate for Arabic than Hindi. In fact, overall results indicate that Arabic accuracy did not significantly differ from English or Dutch. These results join only a few studies which have not found clear in-group advantages in vocal emotion recognition (i.e., McCluskey & Albas, 1981; Zhu, 2013).

A potential explanation for better performance in one unfamiliar language over the other may be explained by indirect exposure and the population demographics of the participants' home country. The Netherlands is home to large numbers of Arabic-speaking immigrant groups (i.e., from Morocco and Syria; ETIAS, 2024). Although our participants may not have had direct experience with or exposure to Arabic (in fact, only 6 participants reported having any level of exposure to Arabic), it is conceivable that our participants may often overhear Arabic conversations in their daily lives. There is ample evidence that passive exposure to a foreign language (without being able to speak or use that language) leads to better auditory perception of that language and can incur lasting changes in brain structure (e.g., Au et al., 2002; Barbeau et al., 2025; Bice & Kroll, 2019; Oh et al., 2010; Pierce et al., 2015). Even indirect exposure to Arabic language (and/or culture, as we discussed above, these two variables are difficult to disentangle) may have contributed to our participants' increased ability to recognize emotions in Arabic prosody. If so, the *language-proficiency advantage hypothesis* supported by our findings may also feature an exposure component, where indirect forms of exposure may be sufficient for high performance in vocal emotion recognition. Future research could more systematically address the effect of indirect language exposure on emotion recognition, for example, by

exposing participants to an unknown language or testing listeners that live in multi-cultural environments and examining changes in emotion recognition ability.

If indirect exposure to Arabic is indeed the driving cause of increased accuracy for Arabic emotion recognition, it is interesting to note that participants did not seem to be aware of this consciously. Again, only 6 participants reported any amount of Arabic exposure. Additionally, while participants tended to rate emotion recognition for both English and Dutch as “somewhat easy” or “extremely easy”, they tended to rate both Arabic and Hindi as “somewhat difficult” or “extremely difficult”. This would indicate that while Dutch-English bilinguals are highly accurate at recognizing emotions in Arabic sentences, subjectively this still felt more difficult than recognizing emotion in English or Dutch. Future research should continue to investigate the relationship between emotion recognition accuracy and other self-reported measures such as perceived confidence or perceived confusions with other emotions.

Limitations and future directions

Aspects of our experimental design and our findings leave some unanswered questions and potential directions for future research. To ensure that differences across the presented languages were not driven by baseline differences in stimuli quality, we incorporated norming values and random effects for items and participants in our generalized linear mixed-effects model. Thus, all reported comparisons attempt to control for potential baseline differences in stimuli quality. Nonetheless, we note that norming rating still explained a certain portion of the observed variance in accuracy, and validity may be limited due to the fact that norming ratings are derived from separate validation studies (i.e., our validation for Dutch stimuli, versus English, Hindi, and Arabic validation from Pell, Paulmann, et al., 2009). Additionally, our experiment included only one speaker of each language. While stimuli were normed by native

speakers of each language and although this format has been used in previous research (i.e., Zhang & Pell, 2022; Pell et al. 2009), it is nonetheless conceivable that potential differences in performance could be influenced by speaker-specific rather than language-specific prosodic features (for examples of studies on emotional prosody that use multiple speakers per language see Banse & Scherer, 1996; Goudbeek & Broersma, 2010; Goudbeek & Scherer, 2010; Laukka et al., 2005; Liang et al., 2025).

Our study may also be limited by the specific languages chosen for comparison to the participants' L1 and L2. Hindi and Arabic were chosen as the two “unknown” languages to facilitate comparisons with previous research who have used these stimuli with monolinguals (i.e., Pell et al., 2009; Zhang & Pell, 2022). However, Hindi and Arabic differ from Dutch and English in more ways than simply being unfamiliar languages—they belong to different language families and vary in linguistic distance from the participants’ known languages (see also Bradlow et al., 2010; Scherer et al., 2001). Future research could more directly investigate these effects by selecting “unknown” languages that share the same language family or have comparable linguistic distance to the known languages.

Lastly, previous work has highlighted alternative perspectives on basic emotion theory and the universality claims in emotion recognition (see Lorette & Dewaele, 2024, for a discussion). Alternative perspectives argue against the “basic” emotion and universality accounts, instead emphasizing the cultural relativity of emotion perception (i.e., Barrett et al., 2011) and providing evidence that emotion recognition is highly context-dependent (i.e., Barrett, 2011; Gendron et al., 2014; Nelson & Russell, 2013). Along these lines, this alternative approach would argue that methodological factors—particularly the use of forced-choice experimental tasks—introduce a particular context that can artificially inflate emotion recognition accuracy,

especially in cross-cultural contexts (i.e., Gendron et al., 2014). Compared to methodologies such as open-ended response formats, forced-choice tasks like ours may prime emotion-concept knowledge, leading to inflated accuracy (Gendron et al., 2014). This particular methodological context (forced-choice task) could be an alternative explanation for the high accuracies we observed, particularly the unexpectedly high recognition observed for Arabic stimuli. Although these perspectives suggest that some universality effects may be influenced by task structure, our findings do align with empirical studies that robustly observe universality effects (e.g., Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Scherer et al., 2001; Thompson & Balkwill, 2006; Zhang & Pell, 2022), demonstrating that prosodic features can be reliably recognized in this particular experimental context. Future research could systematically compare potential task-related factors by including response options that allow participants to indicate uncertainty, such as a “cannot tell” choice, as implemented in previous studies (Lorette & Dewaele, 2019; 2020; 2024), or by using free-response formats (Gendron et al., 2014).

Conclusions

Our results suggest that the in-group advantage in vocal emotion recognition does extend to a highly proficient L2—even for participants living and tested in their L1 cultural environment who have limited experience living in their L2’s native cultural environment. Furthermore, correlations with language proficiency support that L2 emotion recognition can be learned and that this learning corresponds with increases in general language proficiency. These results challenge and extend earlier findings that listeners consistently show strong advantages in emotion recognition for their cultural in-group or native language (i.e., Elfenbein & Ambady, 2002; Paulmann & Uskul, 2014; Pell, Monetta, et al., 2009; Zhang & Pell, 2022). Furthermore, our findings challenge theories such as the theory of cultural learning in emotional behavior and

the dialect theory (Elfenbein, 2013; Laukka et al., 2016) which suggest that in-group advantages in emotion recognition arise from culturally-specific aspects of emotional communication within a cultural community. Instead, our results point to the role of language proficiency, suggesting that high levels of emotion recognition can be achieved with high levels of language proficiency, even without being part of a cultural in-group. Overall, these findings augment the current literature on emotion recognition, and add to the small number of existing studies investigating the role of language proficiency and emotion recognition in bilinguals' two languages. As we argued above, future work is needed to further define the specific contributions of cultural exposure and language proficiency for L2 and unfamiliar language emotion recognition.

670 **Data availability:** The data of this study as well as R code used for data analysis are openly
671 available in OSF at https://osf.io/6ep4w/?view_only=426ebcdc87274f0c8ff85004cdae3c49.
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References

- 674 Alqarni, N., & Dewaele, J.-M. (2020). A bilingual emotional advantage? An investigation into
675 the effects of psychological factors in emotion perception in Arabic and in English of
676 Arabic-English bilinguals and Arabic/English monolinguals. *International Journal of*
677 *Bilingualism*, 24(2), 141–158. <https://doi.org/10.1177/1367006918813597>
- 678 Altrov, R. (2013). Aspects of cultural communication in recognizing emotions. *Trames. Journal*
679 *of the Humanities and Social Sciences*, 17(2), 159. <https://doi.org/10.3176/tr.2013.2.04>
- 680 Au, T. K., Knightly, L. M., Jun, S.-A., & Oh, J. S. (2002). Overhearing a language during
681 childhood. *Psychological Science*, 13(3), 238–243. [https://doi.org/10.1111/1467-](https://doi.org/10.1111/1467-9280.00444)
682 [9280.00444](https://doi.org/10.1111/1467-9280.00444)
- 683 Banse, R., & Scherer, K. R. (1996). Acoustic profiles in vocal emotion expression. *Journal of*
684 *Personality and Social Psychology*, 70(3), 614–636. [https://doi.org/10.1037/0022-](https://doi.org/10.1037/0022-3514.70.3.614)
685 [3514.70.3.614](https://doi.org/10.1037/0022-3514.70.3.614)
- 686 Barbeau, E. B., Pierce, L., Deschamps, S., Kousaie, S., Gilbert, A., Chen, J.-K., Baum, S., &
687 Klein, D. (2025). Early but discontinued exposure to a language exerts lasting effects on
688 white matter architecture in the brain. *Communications Biology*, 8(1), 1575.
689 <https://doi.org/10.1038/s42003-025-08954-4>
- 690 Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for
691 confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*,
692 68(3), 10.1016/j.jml.2012.11.001. <https://doi.org/10.1016/j.jml.2012.11.001>
- 693 Barrett, L. F. (2011). Was Darwin wrong about emotional expressions? *Current Directions*
694 *in Psychological Science*, 20(6), 400–406. <https://doi.org/10.1177/0963721411429125>

695 Barrett, L. F., Mesquita, B., & Gendron, M. (2011). Context in emotion perception. *Current*
696 *Directions in Psychological Science*, 20(5), 286–290.
697 <https://doi.org/10.1177/0963721411422522>

698 Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models
699 using Lme4. *Journal of Statistical Software*, 67, 1–48.
700 <https://doi.org/10.18637/jss.v067.i01>

701 Bhatara, A., Laukka, P., Boll-Avetisyan, N., Granjon, L., Elfenbein, H. A., & Bänziger, T.
702 (2016). Second language ability and emotional prosody perception. *PLOS ONE*, 11(6),
703 e0156855. <https://doi.org/10.1371/journal.pone.0156855>

704 Bice, K., & Kroll, J. F. (2019). English only? Monolinguals in linguistically diverse contexts
705 have an edge in language learning. *Brain and Language*, 196, 104644.
706 <https://doi.org/10.1016/j.bandl.2019.104644>

707 Boersma, P., & Weenink, D. (2022). *Praat: Doing phonetics by computer*.

708 Bradlow, A., Clopper, C., Smiljanic, R., Walter, M.A. (2010). A perceptual phonetic similarity
709 space for languages: Evidence from five native language listener groups. *Speech*
710 *Communication*, 52(11-12), 930-942. doi: 10.1016/j.specom.2010.06.003.

711 Cho, C. M., & Dewaele, J.-M. (2021). A crosslinguistic study of the perception of emotional
712 intonation: Influence of the pitch modulations. *Studies in Second Language Acquisition*,
713 43, 870-895. <https://doi.org/10.1017/S0272263120000674>

714 De Groot, M., Schaafsma, J., Castelain, T., Malinowska, K., Mann, L., Ohtsubo, Y., Wulandari,
715 M. T. A., Bataineh, R. F., Fry, D. P., Goudbeek, M., & Suryani, A. (2021). Group-based
716 shame, guilt, and regret across cultures. *European Journal of Social Psychology*, 51(7),
717 1198–1212. <https://doi.org/10.1002/ejsp.2808>

718 Dromey, C., Silveira, J., & Sandor, P. (2005). Recognition of affective prosody by speakers of
719 English as a first or foreign language. *Speech Communication*, 47(3), 351–359.
720 <https://doi.org/10.1016/j.specom.2004.09.010>

721 Education First. (2024). Netherlands EF English proficiency index. In *Education First*.
722 <https://www.ef.com/wwen/epi/regions/europe/netherlands/>.

723 Edwards, A. (2016). *English in the Netherlands: Functions, forms and attitudes*. University of
724 Cambridge.

725 Ekberg, M., Stavrinou, G., Andin, J., Stenfelt, S., & Dahlström, Ö. (2023). Acoustic features
726 distinguishing emotions in Swedish speech. *Journal of Voice*.
727 <https://doi.org/10.1016/j.jvoice.2023.03.010>

728 Ekman, P., & Friesen, W. V. (1971). Constants across cultures in the face and emotion. *Journal*
729 *of Personality and Social Psychology*, 17(2), 124–129. <https://doi.org/10.1037/h0030377>

730 Elfenbein, H. A. (2013). Nonverbal dialects and accents in facial expressions of emotion.
731 *Emotion Review*, 5(1), 90–96. <https://doi.org/10.1177/1754073912451332>

732 Elfenbein, H. A., & Ambady, N. (2002). On the universality and cultural specificity of emotion
733 recognition: A meta-analysis. *Psychological Bulletin*, 128(2), 203–235.
734 <https://doi.org/10.1037/0033-2909.128.2.203>

735 Elfenbein, H. A., Beaupré, M., Lévesque, M., & Hess, U. (2007). Toward a dialect theory:
736 Cultural differences in the expression and recognition of posed facial expressions.
737 *Emotion (Washington, D.C.)*, 7(1), 131–146. <https://doi.org/10.1037/1528-3542.7.1.131>

738 ETIAS. (2024). Immigration trends shape the Netherlands’ growing population. In *ETIAS*.
739 <https://etias.com/articles/netherlands-population-growth-immigration-trends-2024>.

740 European Commission. (2012). *Europeans and their Languages: Special Eurobarometer 386*.
 741 European Commission.

742 Fox, J., & Weisberg, S. (2019). *An R companion to applied regression*. Sage.

743 Gendron, M., Roberson, D., Van Der Vyver, J. M., & Barrett, L. F. (2014). Cultural relativity in
 744 perceiving emotion from vocalizations. *Psychological Science*, 25(4), 911–920.
 745 <https://doi.org/10.1177/0956797613517239>

746 Goddard, C., & Wierzbicka, A. (2004). Cultural scripts: What are they and what are they good
 747 for? *Intercultural Pragmatics*, 1(2). <https://doi.org/10.1515/iprg.2004.1.2.153>

748 Graham, C. R., Hamblin, A. W., & Feldstein, S. (2001). Recognition of emotion in English
 749 voices by speakers of Japanese, Spanish and English. *IRAL - International Review of*
 750 *Applied Linguistics in Language Teaching*, 39(1). <https://doi.org/10.1515/iral.39.1.19>

751 Goudbeek, M., & Broersma, M. (2010). Language specific effects of emotion on phoneme
 752 duration. *Interspeech 2010*, 2026–2029. <https://doi.org/10.21437/Interspeech.2010-573>

753 Goudbeek, M., & Scherer, K. (2010). Beyond arousal: Valence and potency/control cues in the
 754 vocal expression of emotion. *The Journal of the Acoustical Society of America*, 128(3),
 755 1322–1336. <https://doi.org/10.1121/1.3466853>

756 Hess, U., Thibault, P., & Levesque, M. (2013). Where do emotional dialects come from? A
 757 comparison of the understanding of emotion terms between Gabon and Quebec. In J. R. J.
 758 Fontaine, K. R. Scherer, & C. Soriano (Eds.), *Components of Emotional Meaning* (pp.
 759 512–519). Oxford University Press.
 760 <https://doi.org/10.1093/acprof:oso/9780199592746.003.0044>

761 Ji, R. (2023). Impact of bilingualism on emotion perception. *Journal of Education*, 22.
 762 <https://doi.org/10.54097/ehss.v22i.12434>

763 Juslin, P. N., & Laukka, P. (2003). Communication of emotions in vocal expression and music
 764 performance: Different channels, same code? *Psychological Bulletin*, 129(5), 770–814.
 765 <https://doi.org/10.1037/0033-2909.129.5.770>

766 Larrouy-Maestri, P., Poeppel, D., & Pell, M. D. (2024). The sound of emotional prosody: Nearly
 767 3 decades of research and future directions. *Perspectives on Psychological Science*,
 768 17456916231217722. <https://doi.org/10.1177/17456916231217722>

769 Laukka, P., & Elfenbein, H. A. (2021). Cross-cultural emotion recognition and in-group
 770 advantage in vocal expression: A meta-analysis. *Emotion Review*, 13(1), 3–11.
 771 <https://doi.org/10.1177/1754073919897295>

772 Laukka, P., Elfenbein, H. A., Thingujam, N. S., Rockstuhl, T., Iraki, F. K., Chui, W., & Althoff,
 773 J. (2016). The expression and recognition of emotions in the voice across five nations: A
 774 lens model analysis based on acoustic features. *Journal of Personality and Social*
 775 *Psychology*, 111(5), 686–705. <https://doi.org/10.1037/pspi0000066>

776 Laukka, P., Juslin, P., & Bresin, R. (2005). A dimensional approach to vocal expression of
 777 emotion. *Cognition & Emotion*, 19(5), 633–653.
 778 <https://doi.org/10.1080/02699930441000445>

779 Lemhöfer, K., & Broersma, M. (2012). Introducing LexTALE: A quick and valid lexical test for
 780 advanced learners of English. *Behavior Research Methods*, 44(2), 325–343.
 781 <https://doi.org/10.3758/s13428-011-0146-0>

782 Lenth, R. V. (2024). *Emmeans: Estimated Marginal Means, aka Least-Squares Means*.

783 Liang, Y., Goudbeek, M., Konopka, A., Choi, J., & Broersma, M. (2025). Investigating cross-
 784 cultural vocal emotion recognition with an affectively and linguistically balanced design.

785 *Language and Speech*, 00238309251318730.

786 <https://doi.org/10.1177/00238309251318730>

787 Lorette, P., & Dewaele, J.-M. (2015). Emotion recognition ability in English among L1 and LX

788 users of English. *International Journal of Language and Culture*, 2(1), 62–86.

789 <https://doi.org/10.1075/ijolc.2.1.03lor>

790 Lorette, P., & Dewaele, J.-M. (2019). The relationship between bi/multilingualism,

791 nativeness, proficiency and multimodal emotion recognition ability. *International Journal*

792 *of Bilingualism*, 23(6), 1502–1516. <https://doi.org/10.1177/1367006918808042>

793 Lorette, P., & Dewaele, J.-M. (2020). Emotion recognition ability across different

794 modalities: The role of language status (L1/LX), proficiency and cultural background.

795 *Applied Linguistics Review*, 11(1), 1–26. <https://doi.org/10.1515/applirev-2017-0015>

796 Lorette, P., & Dewaele, J.-M. (2024b). Valence and arousal perception among first language

797 users, foreign language users, and naïve listeners of Mandarin across various

798 communication modalities. *Bilingualism: Language and Cognition*, 27(5), 874–885.

799 <https://doi.org/10.1017/S1366728923000925>

800 McCluskey, K. W., & Albas, D. C. (1981). Perception of the emotional content of speech by

801 Canadian and Mexican children, adolescents, and adults. *International Journal of*

802 *Psychology*, 16(2), 119–132. <https://doi.org/10.1080/00207598108247409>

803 Mesquita, B., Boiger, M., & De Leersnyder, J. (2017). Doing emotions: The role of culture in

804 everyday emotions. *European Review of Social Psychology*, 28(1).

805 <https://doi.org/10.1080/10463283.2017.1329107>

806 Nelson, N. L., & Russell, J. A. (2013). Universality revisited. *Emotion Review*, 5(1), 8–15.

807 <https://doi.org/10.1177/1754073912457227>

808 Oh, J. S., Au, T. K.-F., & Jun, S.-A. (2010). Early childhood language memory in the speech
809 perception of international adoptees. *Journal of Child Language*, 37(5), 1123–1132.
810 <https://doi.org/10.1017/S0305000909990286>

811 Paulmann, S., & Uskul, A. K. (2014). Cross-cultural emotional prosody recognition: Evidence
812 from Chinese and British listeners. *Cognition & Emotion*, 28(2), 230–244.
813 <https://doi.org/10.1080/02699931.2013.812033>

814 Pell, M. D., & Kotz, S. A. (2011). On the time course of vocal emotion recognition. *PLoS ONE*,
815 6(11), e27256. <https://doi.org/10.1371/journal.pone.0027256>

816 Pell, M. D., Monetta, L., Paulmann, S., & Kotz, S. A. (2009). Recognizing emotions in a foreign
817 language. *Journal of Nonverbal Behavior*, 33(2), 107–120.
818 <https://doi.org/10.1007/s10919-008-0065-7>

819 Pell, M. D., Paulmann, S., Dara, C., Allasseri, A., & Kotz, S. A. (2009). Factors in the recognition
820 of vocally expressed emotions: A comparison of four languages. *Journal of Phonetics*,
821 37(4), 417–435. <https://doi.org/10.1016/j.wocn.2009.07.005>

822 Pierce, L. J., Chen, J.-K., Delcenserie, A., Genesee, F., & Klein, D. (2015). Past experience
823 shapes ongoing neural patterns for language. *Nature Communications*, 6(1), 10073.
824 <https://doi.org/10.1038/ncomms10073>

825 Sauter, D. A., Eisner, F., Calder, A. J., & Scott, S. K. (2010). Perceptual cues in nonverbal vocal
826 expressions of emotion. *Quarterly Journal of Experimental Psychology*, 63(11), 2251–
827 2272. <https://doi.org/10.1080/17470211003721642>

828 Sauter, D. A., Eisner, F., Ekman, P., & Scott, S. K. (2010). Cross-cultural recognition of basic
829 emotions through nonverbal emotional vocalizations. *Proceedings of the National*
830 *Academy of Sciences*, 107(6), 2408–2412. <https://doi.org/10.1073/pnas.0908239106>

831 Scherer, K. R. (2003). Vocal communication of emotion: A review of research paradigms.
832 *Speech Communication*, 40(1), 227–256. [https://doi.org/10.1016/S0167-6393\(02\)00084-5](https://doi.org/10.1016/S0167-6393(02)00084-5)

833 Scherer, K. R., Banse, R., & Wallbott, H. G. (2001). Emotion inferences from vocal expression
834 correlate across languages and cultures. *Journal of Cross-Cultural Psychology*, 32(1),
835 76–92. <https://doi.org/10.1177/0022022101032001009>

836 Schutte, N. S., Malouff, J. M., Hall, L. E., Haggerty, D. J., Cooper, J. T., Golden, C. J., &
837 Dornheim, L. (1998). Development and validation of a measure of emotional
838 intelligence. *Personality and Individual Differences*, 25(2), 167–177.
839 [https://doi.org/10.1016/S0191-8869\(98\)00001-4](https://doi.org/10.1016/S0191-8869(98)00001-4)

840 Thompson, W. F., & Balkwill, L.-L. (2006). Decoding speech prosody in five languages.
841 *Semiotica*, 158, 407–424. <https://doi.org/10.1515/SEM.2006.017>

842 Yildirim, S., Bulut, M., Lee, C. M., Kazemzadeh, A., Deng, Z., Lee, S., Narayanan, S., & Busso,
843 C. (2004). An acoustic study of emotions expressed in speech. *Interspeech 2004*, 2193–
844 2196. <https://doi.org/10.21437/Interspeech.2004-242>

845 Zhang, S., & Pell, M. D. (2022). Cultural differences in vocal expression analysis: Effects of
846 task, language, and stimulus-related factors. *PloS One*, 17(10), e0275915.
847 <https://doi.org/10.1371/journal.pone.0275915>

848 Zhu, Y. (2013). Which is the best listener group?: Perception of Chinese emotional prosody by
849 Chinese natives, naïve Dutch listeners and Dutch L2 learners of Chinese. *Dutch Journal*
850 *of Applied Linguistics*, 2(2), 170–183. <https://doi.org/10.1075/dujal.2.2.03zhu>

851

852 **Table 1**

853 *Average percentage (SD) of native speakers who correctly identified the emotion of the final*
 854 *stimuli in their respective validation studies (English, Hindi, and Arabic from Pell, Paulmann, et*
 855 *al., 2009).*

Emotion	Language			
	Dutch	English	Arabic	Hindi
Angry	95.00% (3.47%)	71.11% (8.37%)	78.28% (6.27%)	89.17% (5.72%)
Fearful	83.06% (2.53%)	89.21% (4.31%)	58.82% (12.49%)	42.92% (5.94%)
Sad	93.33% (3.60%)	97.92% (2.09%)	69.61% (10.70%)	96.25% (4.15%)
Happy	65.28% (3.96%)	95.49% (3.17%)	50.00% (11.90%)	86.67% (4.72%)
Neutral	87.22% (4.48%)	89.86% (3.61%)	70.13% (8.14%)	20.42% (6.61%)

856

857 **Table 2**

858 *Participants' subjective difficulty ratings for recognizing Dutch, English, Arabic, and Hindi*
 859 *emotions and overall emotion recognition difficulty (% of participants).*

Language	Response				
	Extremely easy	Somewhat easy	Neither easy nor difficult	Somewhat difficult	Extremely difficult
Dutch	22.92%	54.17%	6.25%	12.5%	4.17%
English	20.83%	54.17%	10.42%	10.42%	4.17%
Arabic	4.17%	16.67%	14.58%	50%	14.58%
Hindi	0%	20.83%	12.5%	35.42%	31.25%
Overall Task Difficulty	0%	27.08%	18.75%	54.17%	0%

860

Figure 1

Confusion matrix describing participant responses (%) for each language and emotion stimuli condition. Cells on the diagonal represent accurate emotion recognition; all other cells represent inaccurate recognition.

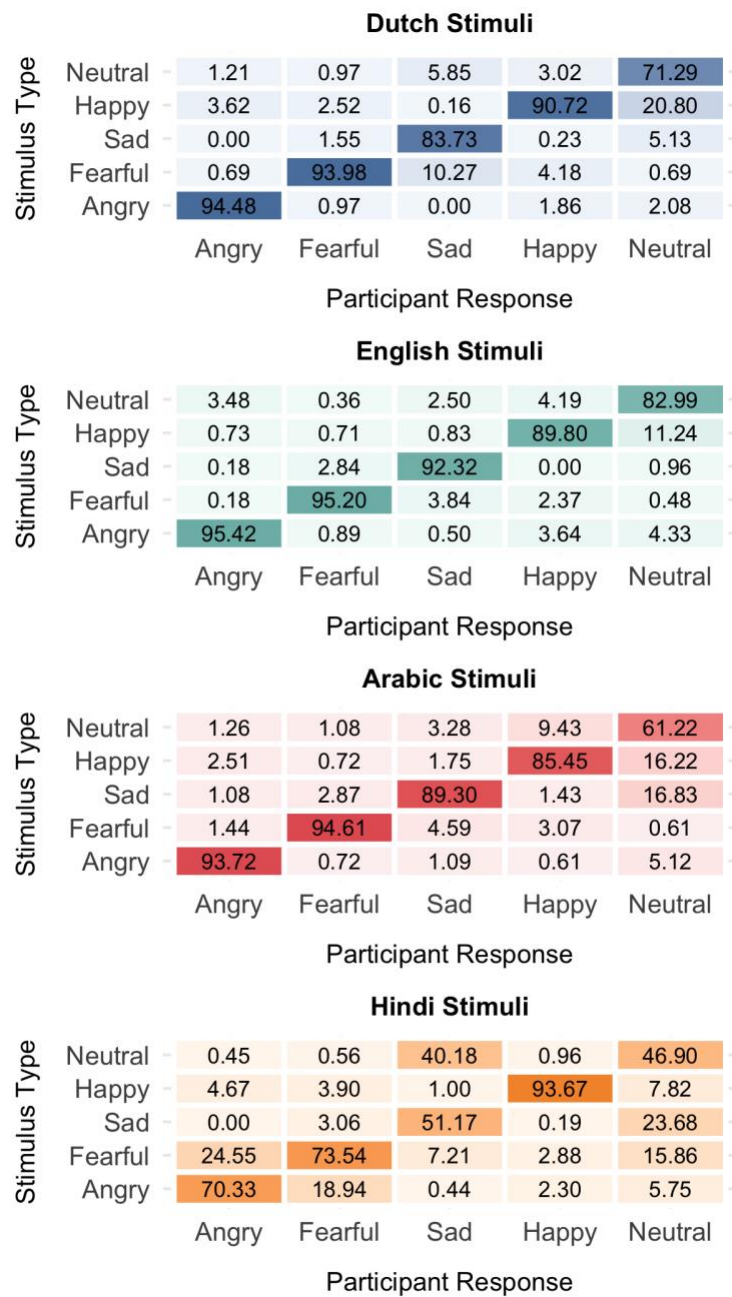


Figure 2

*Predicted emotion recognition accuracies by language and emotion type. Error bars represent 95% confidence intervals. Overall mean for each language indicated by black diamond, with bars indicating significant differences (** for $p < .01$ and *** for $p < .001$). Dotted line represents chance performance at 20%.*

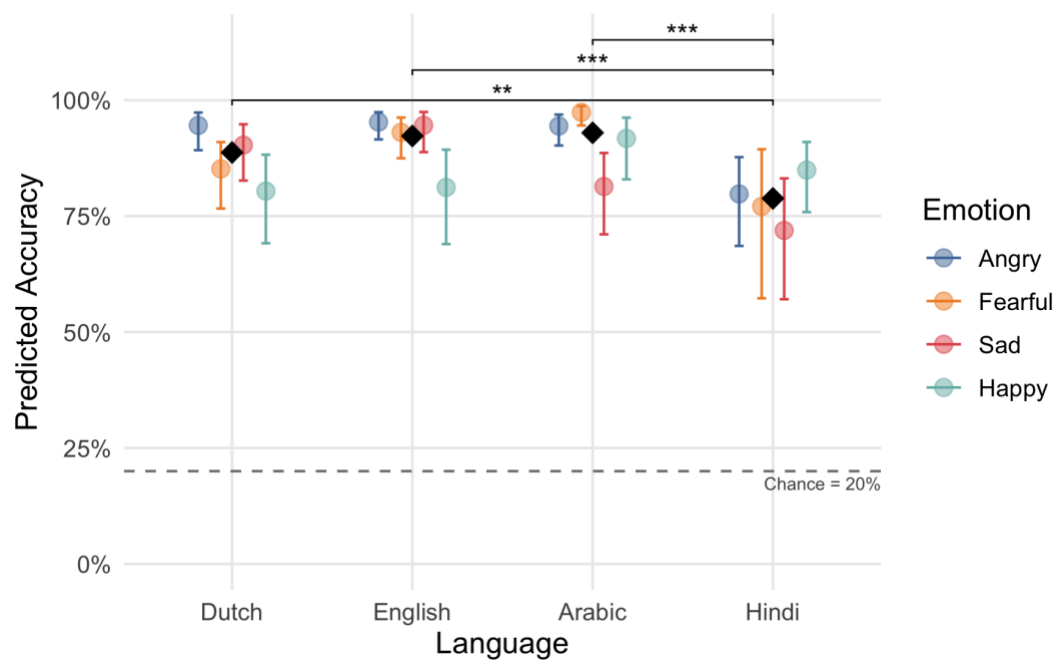


Figure 3

Pearson's correlations between Dutch and English emotion recognition accuracy and language proficiency measures including: self-rated Dutch proficiency, self-rated English proficiency, English LexTALE score, and English age of acquisition (AoA). * $p_{adj} < .05$, ** $p_{adj} < .01$, *** $p_{adj} < .001$

